Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing (with Business Management)**

Unit ID	**2232**
Assessment	**Computer Networks 2.0**
Assessment Type	**Global Examination**
Date	**Wednesday 4th June 2025**
Time	**09.00AM – 12.00PM**

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Explain the primary purpose of the Open Systems Interconnection (OSI) seven-layer model in defining network architecture **and** outline its key importance in modern network communication, including its role in standardising protocols, enabling interoperability, simplifying troubleshooting, and facilitating modular network design. **4**

b) Identify and describe the primary functions of **TWO (2) specific** OSI layers (excluding Layer 4, the Transport Layer), and for each layer, provide examples of protocols that operate at that layer, explaining their role in network communication. **6**

Total 10 marks

Question 2

a) Describe the primary roles and functional characteristics of the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) in network communication, including their key differences in reliability, connection management, data transfer mechanisms, and typical use cases. **5**

b) Explain the primary purpose of the Hypertext Transfer Protocol (HTTP) in client-server communication and describe in detail how it facilitates interaction between clients (e.g., web browsers) and web servers, including key mechanisms such as request-response cycles, message formats, and statelessness. **5**

Total 10 marks

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Question 3

a) Explain the primary purpose of subnetting in computer networks and outline specific ways in which it improves network efficiency, including examples of its impact on traffic management, address utilisation, and scalability. **5**

b) Explain the key features of star and mesh network topologies and discuss their relative advantages. **5**

Total 10 marks

Question 4

a) Compare the key features and performance differences between the IEEE 802.11a, 802.11b, 802.11g, 802.11n (Wi-Fi 4), and 802.11ac (Wi-Fi 5) wireless standards. Address the following aspects for each standard: **5**

- Frequency band(s) used
- Maximum theoretical data rate
- Channel width
- Modulation techniques
- Spatial stream support (MIMO)
- Backward compatibility
- Typical use cases and era of prevalence

b) Recommend the most suitable IEEE 802.11 wireless standard for a busy open-plan office environment with high device density and frequent data transfers (e.g., video conferencing, cloud storage). Justify your choice by evaluating key performance factors such as frequency bands, channel capacity, interference resistance, data throughput, and support for modern features like MU-MIMO and OFDMA. **5**

Total 10 marks

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Question 5

a) Explain the primary purpose of a firewall in network security and detail its operational mechanisms for protecting a network. Address the following: **5**

- Core objectives (e.g., traffic filtering, access control, threat mitigation)
- Key types of firewalls (e.g., packet-filtering, stateful inspection, application-level gateway)
- Criteria used to filter traffic (e.g., IP addresses, ports, protocols, application layer data)
- Role in preventing unauthorised access and common cyber threats (e.g., malware, brute-force attacks)

b) Describe how Virtual Local Area Networks (VLANs) **and** Access Control Lists (ACLs) contribute to internal network security. Specifically address: **5**

- The role of VLANs in segmenting network traffic, isolating sensitive data, and limiting lateral movement of threats.
- The function of ACLs in enforcing granular traffic filtering based on criteria like IP addresses, ports, protocols, and traffic direction.
- How the combination of VLANs and ACLs strengthens security by reducing attack surfaces, controlling inter-VLAN communication, and preventing unauthorised access between network segments.

Total 10 marks

Question 6

a) Describe the core principle of network segmentation in business networks **and** explain how it enhances both security and performance. Address the following: **4**

- Fundamental concept of dividing a network into logical segments (e.g., VLANs, subnets).
- Specific security benefits (e.g., limiting lateral movement of threats, isolating sensitive data, reducing attack surfaces).
- Performance improvements (e.g., traffic optimisation, reduced broadcast domains, efficient resource allocation).
- Practical examples of segmentation strategies (e.g., separating guest networks, isolating IoT devices, segmenting departments).

b) Describe how each of the following commands or protocol is used in network fault diagnosis **and** how each assist in resolving network issues. **6**

- Ping
- Traceroute
- SNMP

Total 10 marks

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Question 7

a) Describe the primary functions of the Dynamic Host Configuration Protocol (DHCP) and the Domain Name System (DNS) in a network environment. For each, address: **5**

- Core purpose and role in network operation
- Technical mechanisms or processes they employ
- How they enable connectivity and usability for end-users
- Interactions with other network components (e.g., routers, switches, client devices)

b) Explain how server software supports essential network services, including file sharing, print services, and centralised management. For each service, address: **5**

- Core functionality provided by server software
- Technical protocols or mechanisms used (e.g., SMB, NFS, CUPS, LDAP)
- How server software enables secure, efficient delivery of the service
- Role in facilitating centralised control and management of network resources

Total 10 marks

Question 8

a) Explain the key benefits and limitations of using Voice over Internet Protocol (VoIP) in a modern office network. Address the following for each category: **5**

- Benefits
  - Cost-efficiency (e.g., reduced infrastructure, long-distance savings)
  - Functionality (e.g., unified communications, scalability, advanced features like video conferencing)
  - Integration capabilities (e.g., with CRM systems, cloud services)
  - Flexibility for remote work/hybrid environments
- Limitations
  - Dependence on network infrastructure (e.g., bandwidth requirements, latency/quality issues)
  - Security vulnerabilities (e.g., eavesdropping, denial-of-service attacks)
  - Reliance on power/Internet connectivity
  - Potential compatibility challenges with legacy systems
- Practical considerations for balancing benefits and limitations in office deployments (e.g., QoS implementations, security protocols, redundancy planning)

b) Describe how a video conferencing system can be integrated into an existing LAN setup. **5**

Total 10 marks

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Question 9

a) What is a Virtual Private Network (VPN), and how does it secure remote communication for employees? Address the following: **5**

1. Core Definition
2. Security Mechanisms
3. Protection of Remote Communication
4. Role in Remote Work Security
5. Practical Components in Employee Use Cases

b) Describe a specific corporate scenario where implementing Virtual Private Network (VPN) access control policies would significantly enhance security. In your description, include the following details: **5**

1. The Business Context
2. The Security Threats
3. The VPN Access Control Policies
4. The Security Improvements

Total 10 marks

Question 10

a) Propose a comprehensive network monitoring strategy that leverages Simple Network Management Protocol (SNMP) and log analysis tools. In your proposal, address the following aspects: **5**

- Objectives and Scope
- SNMP Implementation
- Log Analysis
- Alerting and Response
- Tools and Technologies
- Scalability and Maintenance

b) Explain, with specific examples, how maintaining up-to-date network documentation, including network diagrams and change logs, supports network troubleshooting and management processes. In your explanation, address the following aspects: **5**

- Role in Troubleshooting
- Benefits for Network Management
- Impact on Efficiency

Total 10 marks

End of paper

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